LINUX TROUBLESHOOTING GUIDE

A beginner-friendly reference for IT Helpdesk and RAG chatbot testing

Topics: Linux basics • terminal commands • files and permissions • SSH • services • disk space • logs • networking

Purpose: Use this guide as a knowledge-base document for your AI IT Helpdesk chatbot. The examples are designed to be searchable and easy for a retrieval-augmented generation (RAG) app to answer from.

Practice safely: Commands that change files, restart services, or modify access can affect a real system. Run practice commands in a test environment and ask your administrator before changing production systems.

1. What is Linux?

Linux is an operating-system kernel used in servers, cloud platforms, desktops, and many embedded devices. Linux distributions such as Ubuntu, Debian, and Red Hat Enterprise Linux (RHEL) package the kernel with tools and applications to provide a complete operating system.

Why IT support teams use it: Linux servers commonly host websites, applications, databases, and background services. Helpdesk engineers use terminal commands to inspect system health, troubleshoot connectivity, review logs, and manage services.

What is the terminal? The terminal lets you type commands to interact with the operating system. A shell such as Bash reads and runs those commands.

2. Essential Linux commands

Use these commands first when investigating a problem. Most are read-only and suitable for initial checks.

pwd — show the current directory

```
pwd
```

ls -lah — list files, including hidden files, with readable sizes

```
ls -lah
```

cd /var/log — move to the log directory

```
cd /var/log
```

whoami — show the current username

```
whoami
```

hostname — show the system hostname

```
hostname
```

uname -a — show kernel and system information

```
uname -a
```

uptime — show how long the system has been running and load averages

```
uptime
```

date — show the system date and time

```
date
```

3. Files and directories

Linux uses a single directory tree starting at /. Common locations include /home (user files), /etc (system configuration), /var/log (many system logs), /tmp (temporary files), and /var (variable application data). Exact layouts can differ by distribution and software.

Check the size of a directory

```
du -sh /var/log
```

Check filesystem space

```
df -h
```

Find a file by name under a chosen directory

```
find /var/log -type f -name "*.log"
```

Example: If an application reports “disk full,” run df -h to see which mounted filesystem is nearly full. Then investigate large directories with du -sh on relevant paths. Do not delete logs or application files until you understand the impact and follow your team's retention policy.

4. File permissions

Linux permissions control who can read, write, or execute a file. The three permission groups are owner, group, and others. In a listing such as -rw-r-----, the owner can read/write, the group can read, and others have no access.

Inspect ownership and permissions

```
ls -l /path/to/file
```

Check your user and group memberships

```
id
```

Example: “Permission denied” may mean your account lacks the required permission, the parent directory cannot be traversed, or a security policy blocks access. Check the exact path, owner, group, and requested operation before asking an administrator to make a change.

Caution: Avoid using chmod 777 as a quick fix. It grants broad permissions and can create a security risk. Change permissions only to the minimum required, with authorization.

5. SSH connection troubleshooting

SSH (Secure Shell) is commonly used to connect securely to a remote Linux host. A connection failure can be caused by a wrong hostname, DNS issue, network route, firewall, unavailable SSH service, incorrect username, or authentication/key configuration.

Step 1 — confirm the destination name resolves

```
getent hosts server.example.com
```

Step 2 — check basic network reachability (if ICMP is allowed)

```
ping -c 4 server.example.com
```

Step 3 — test whether the SSH port is reachable, if nc is installed

```
nc -vz server.example.com 22
```

Step 4 — connect with verbose client output

```
ssh -v username@server.example.com
```

Common SSH messages:

“Could not resolve hostname” — check spelling, DNS, and whether the host is on the expected network or VPN.

“Connection timed out” — check routing, firewall rules, VPN, destination address, and whether the server is reachable.

“Connection refused” — the host may be reachable but nothing is accepting connections on that port, or a firewall actively rejects it. Verify the SSH service and port with the server administrator.

“Permission denied (publickey)” — verify the username, that the correct private key is selected, and that the matching public key is authorized for that account. Never share a private key.

Example: If SSH returns “Permission denied (publickey),” first confirm you are using the correct username and key. Use ssh -v to identify which key the client offers. Do not copy private keys into tickets or chat messages.

6. Services and processes

A service is a background program, such as a web server or SSH daemon. Many modern distributions use systemd and the systemctl command. Some commands require administrator privileges; do not restart production services without approval.

Check a service status

```
systemctl status sshd
```

On some Debian/Ubuntu systems, the service may be named ssh instead:

```
systemctl status ssh
```

List failed systemd services

```
systemctl --failed
```

View running processes

```
ps aux
```

Check memory usage

```
free -h
```

Troubleshooting example: If an application is unavailable, check its service status and recent logs first. Capture the error and timestamp. Restart only after understanding the impact and following the change process.

7. Logs and error investigation

Logs record system and application events. Start with the relevant service log and the time the problem occurred. Log locations vary by application and Linux distribution.

View recent system journal entries

```
journalctl -n 100 --no-pager
```

View logs for a specific service

```
journalctl -u sshd --since "1 hour ago"
```

Follow a log as new entries arrive

```
tail -f /var/log/syslog
```

On some systems, common log files may instead include /var/log/messages or application-specific paths. If a file does not exist, use journalctl or consult the system's documentation.

Good ticket notes include: exact error text, hostname or affected service, timestamp and timezone, commands run, relevant output, scope of impact, and actions already taken. Remove secrets and personal data before sharing logs.

8. Basic network troubleshooting

Start with the simplest checks and work outward: local interface, IP configuration, default route, DNS resolution, and destination port. A failed ping alone does not prove a server is down because ICMP may be blocked.

Show IP addresses and network interfaces

```
ip addr
```

Show routing table

```
ip route
```

Check DNS resolution

```
getent hosts example.com
```

Test HTTPS connectivity (if curl is installed)

```
curl -I --max-time 10 https://example.com
```

DNS problem: If a hostname fails but a known approved IP is reachable, investigate DNS configuration and name resolution. Do not hard-code an IP as a permanent workaround without approval.

Application port problem: If the server responds but an application cannot connect, check the correct hostname, port, service status, network policy, and application logs.

9. Quick troubleshooting checklist

1. Clarify the symptom and when it started.
2. Identify the affected host, user, application, and business impact.
3. Check whether the issue affects one user or many.
4. Run read-only diagnostic commands first.
5. Review logs around the incident time.
6. Compare with a known-good host or recent change, if allowed.
7. Record evidence and escalate with clear findings.
8. Make changes only with authorization and an appropriate rollback plan.

10. Practice questions for your AI chatbot

Use these questions after uploading this PDF to the chatbot. Good RAG answers should be grounded in the document and mention relevant commands.

Q: What is Linux?
Expected answer: Linux is an operating-system kernel used in distributions such as Ubuntu and RHEL; it is common on servers and cloud platforms.

Q: How do I check disk space?
Expected answer: Run df -h to inspect filesystem usage. Use du -sh on relevant directories to investigate, and do not delete files without understanding the impact.

Q: What does “Permission denied (publickey)” mean?
Expected answer: SSH public-key authentication failed. Check username, selected private key, and whether the matching public key is authorized; never share the private key.

Q: How do I view recent logs for a service?
Expected answer: Use journalctl -u service-name and optionally add a time filter such as --since "1 hour ago".

Q: Why might a hostname fail to connect?
Expected answer: Possible causes include spelling, DNS, network/VPN, routing, firewall, service availability, or authentication. Diagnose based on the exact error.

11. Important safety reminders

• Prefer diagnostic, read-only commands before changing anything.
• Never paste passwords, tokens, private SSH keys, or sensitive logs into a chatbot.
• Do not run commands copied from an untrusted source without understanding them.
• Avoid deleting files, changing permissions, editing firewall rules, or restarting production services without approval.
• Treat chatbot answers as guidance; verify important steps against your organization's runbooks.

End of guide — designed as a starter knowledge-base PDF for an AI IT Helpdesk RAG project.